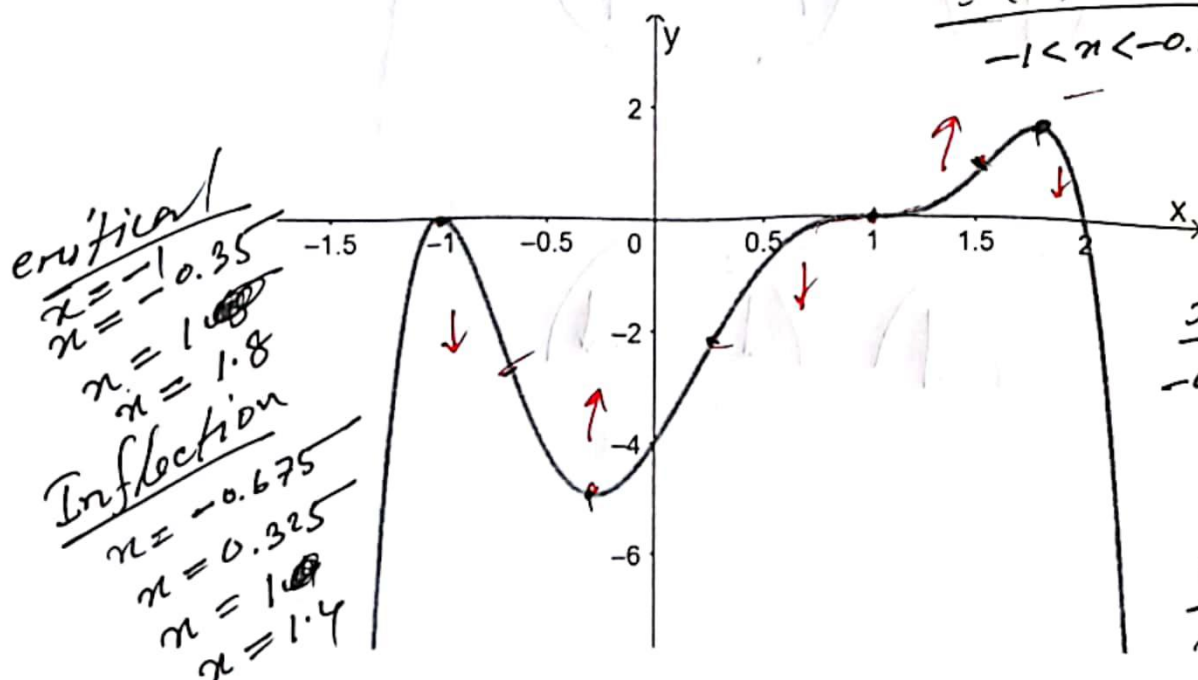


$$8 - 12 + 2 = -2$$

Math-2183 Section-C Class test-01

1. According to the following figure find:

- Find the signs of $f'(x)$ and $f''(x)$. [3]
- What are the critical and inflection points? [2]



$$f'(x) > 0$$

$$x < -1, -0.35 < x < 1.8$$

$$f'(x) < 0$$

$$-1 < x < -0.35, x > 1.8$$

$$f''(x) > 0$$

$$-0.675 < x < 0.325$$

$$1 < x < 1.4$$

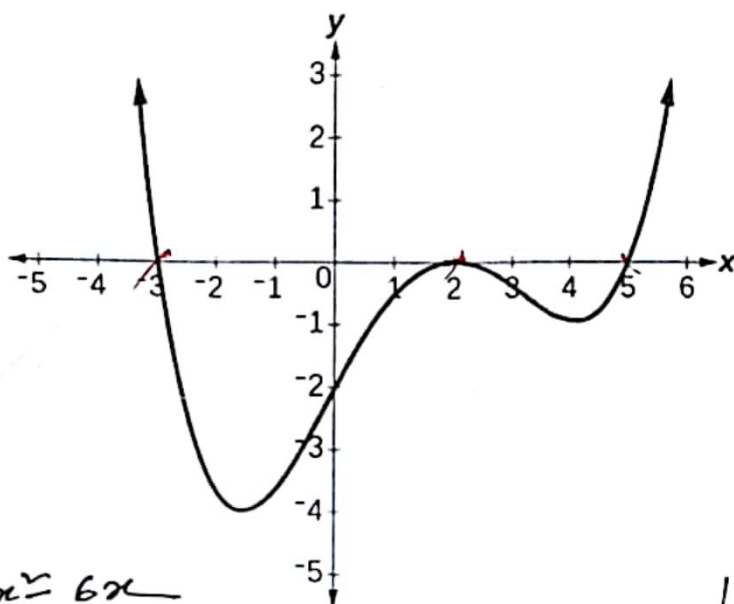
$$f''(x) < 0$$

$$x < -0.675$$

$$0.325 < x < 1.4$$

2. Consider $f(x) = x^3 - 3x^2 + 2$ to find the maximum and minimum by the first derivative test. Also, verify by the second derivative. [3]

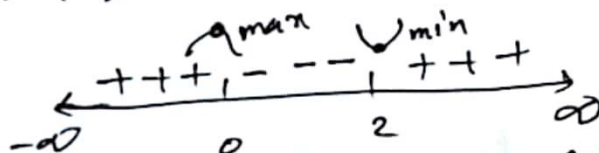
3. According to the following figure, find the nature (even/odd multiplicities) of the roots. [2]



Roots at $x = -3, 2, 5$
odd multiplicity
even multiplicity

$$f'(x) = 3x^2 - 6x$$

$$f'(x) = 0 \Rightarrow 3x(x - 2) = 0 \Rightarrow x = 0, 2$$



$$f_{\max}(0) = 2$$

$$f_{\min}(2) = -2$$

$$f''(x) = 6x - 6$$

$$f''(0) = -6 < 0$$

$$f''(2) = 6 > 0$$

$$\text{max}$$

$$\text{min}$$

Increase $-1.922 < x < -0.56$, ~~$x > 0.232$~~
 Decrease $x < -1.922$, $-0.56 < x < 0.232$

Math-2183 Section-B Class test-01

critical points $x = -1.922$
 $x = -0.56$
 $x = 0.232$

1. According to the following figure find:

- The intervals at which the graph of function increases/decreases. [2]
- Critical points. [1]
- Maximum/minimum values of the given function. [1]
- The intervals at which the graph of function concave up/down. [2]
- Inflection points. [1]

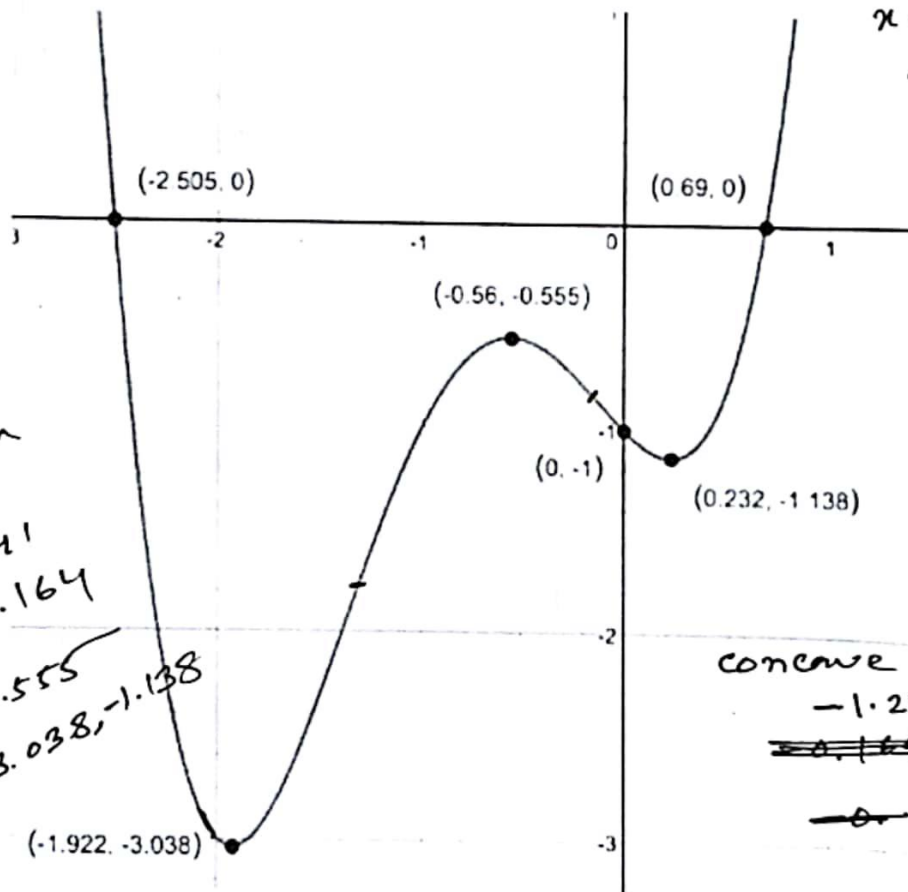
concave up

$x < -1.241$
 $x > -0.164$

Inflection points

$x = -1.241$
 $x = -0.164$

$f_{max} = -0.555$
 $f_{min} = -3.038, -1.138$



concave down
 $-1.241 < x < -0.164$
 ~~$-0.164 < x < 0$~~
 ~~$-0.164 < x < 0$~~

2. For the polynomial $f(x) = x^4(x^2 - 4)^3(x - 1)$ find the nature of the graph at the position of the roots. [2]

3. Find the point at which $f(x) = \frac{1}{2}x^2$ changes its shape. [1]

Roots $x = 0$ multiplicity 4 (not cross x-axis)
 $x = \pm 2$ " 3 (cross x-axis)
 $x = 1$ " 1 (cross x-axis)

$$f(x) = \frac{1}{2}x^2 \rightarrow f'(x) = x \rightarrow f''(x) = 1 \neq 0$$

NO change in ~~shape~~ shape.